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Application of species distribution models to explain and predict the distribution, abundance and assemblage structure of nearshore temperate reef fishes

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ABSTRACT

Aim The purpose of this study was to create predictive species distribution models (SDMs) for temperate reef-associated fish species densities and fish assemblage diversity and richness to aid in marine conservation and spatial planning.

Location California, USA.

Methods Using generalized additive models, we associated fish species densities and assemblage characteristics with seafloor structure, giant kelp biomass and wave climate and used these associations to predict the distribution and assemblage structure across the study area. We tested the accuracy of these predicted extrapolations using an independent data set. The SDMs were also used to estimate larger scale abundances to compare with other estimates of species abundance (uniform density extrapolation over rocky reef and density extrapolations taking into account variations in geomorphic structure).

Results The SDMs successfully modelled the species–habitat relationships of seven rocky reef-associated fish species and showed that species' densities differed in their relationships with environmental variables. The predictive accuracy of the SDMs ranged from 0.26 to 0.60 (Pearson's *r* correlation between observed and predicted density values). The SDMs created for the fish assemblage-level variables had higher prediction accuracies with Pearson's *r* values of 0.61 for diversity and 0.71 for richness. The comparisons of the different methods for extrapolating species densities over a single marine protected area varied greatly in their abundance estimates with the uniform extrapolation (density values extrapolated evenly over the rocky reef) always estimating much greater abundances. The other two methods, which took into account variation in the geomorphic structure of the reef, provided much lower abundance estimates.

Main conclusions Species distribution models that combine geomorphic, oceanographic and biogenic habitat variables can reliably predict spatial patterns of species density and assemblage attributes of temperate reef fishes at spatial scales of 50 m. Thus, SDMs show great promise for informing spatial and ecosystem-based approaches to conservation and fisheries management.

Keywords

generalized additive models, marine landscape ecology, marine protected areas, marine spatial management, species distribution models, temperate reef fishes

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INTRODUCTION

As the fields of resource management and conservation continue to move towards more spatial and ecosystem-based approaches (e.g. terrestrial reserves, marine protected areas), there is a growing need for spatially explicit, quantitative information on species distributions and a better understanding of biotic and abiotic determinants of those distributions (Franklin, 1993; Austin, 1998; Elith & Burgman, 2002; Hamilton *et al.*, 2010; Carr *et al.*, 2011). Landscape approaches, including species distribution models (SDMs), that discriminate habitat types and quantify species–habitat associations at broad geographic scales have become critical to increasing our understanding of how and why populations are distributed throughout their environment to better inform spatial ecosystem-based management (Guisan & Zimmermann, 2000; Hirzel & Guisan, 2002). While the need and application of landscape ecology for informing spatial and ecosystem-based management have long been recognized by managers and conservation scientists in terrestrial environments (Saunders *et al.*, 1991; Franklin, 1993), it has more recently been recognized in the marine environment (Sala *et al.*, 2002; Appeldoorn *et al.*, 2003; Monaco *et al.*, 2005; Friedlander *et al.*, 2007; Pittman *et al.*, 2007; Moore *et al.*, 2011). Of particular interest in marine ecosystem-based management is the growing application of marine protected areas (MPAs) and coastal and marine spatial planning (Ray, 1999; Roberts *et al.*, 2003; Crowder & Norse, 2008; Douvère, 2008; Halpern *et al.*, 2008; Foley *et al.*, 2010; Grorud-Colvert *et al.*, 2011, 2014).

In temperate marine environments, numerous studies have detected strong spatial associations between, for example, temperate demersal fishes and either biotic (e.g. Choat & Ayling, 1987; Levin & Hay, 1996; Dean *et al.*, 2000; Anderson & Millar, 2004) or abiotic (e.g. Carr, 1991; Curley *et al.*, 2002; Harman *et al.*, 2003) environmental variables. Unfortunately, many of these descriptions of species–habitat associations are strictly qualitative and do not allow for making quantitative predictions of species densities beyond the areas sampled (Chatfield *et al.*, 2010). Moreover, the application of SDMs in marine ecosystems has rarely combined high-resolution geomorphic, oceanographic and biogenic variables to predict species density or community attributes at small spatial scales. Here, we construct SDMs that integrate high-resolution bathymetric data, wave exposure and biogenic habitat (kelp biomass) with *in situ* species surveys to (1) characterize benthic habitats, (2) determine species–habitat associations, (3) generate species density distributions across coastal seafloor landscapes, and (5) estimate species abundances. We apply this approach on several species of nearshore fishes along the central coast of California. We focus on central California because of the confluence of sources of information (habitat maps, ecological surveys) required to develop SDMs and the need for this information to inform management of ‘data-poor’ fisheries and the evaluation and adaptive management of a recently established network of MPAs.

The state of California, USA, recently (2007–2012) established a state-wide network of MPAs along the 1200-km coastline (13,688 km² of state waters, 2207 km² or 16% in MPAs; <http://www.dfg.ca.gov/mlpa/>). In response, ecological and oceanographic monitoring programs were implemented throughout the state to assess MPA performance. The combination of MPAs and monitoring programs provides opportunities to scientists and managers for advancing marine ecosystem-based management approaches (Carr *et al.*, 2011; Gleason *et al.*, 2013; Kirlin *et al.*, 2013; Botsford *et al.*, 2014).

Recognizing the need for more and better information on the structure of the seafloor to inform ecosystem-based management, including the design of California’s network of MPAs, the state simultaneously designed and implemented a state-wide seafloor mapping project (California Seafloor Mapping Program (CSMP)). Seafloor mapping is becoming a common and first step for fisheries management because of the close relationship between species distributions and variations in seafloor structure (Kostylev *et al.*, 2001). For example, complex seafloors can provide microhabitats for fish species that serve as refuge from predation (Scharf *et al.*, 2006) and increase species’ abundances (Walker *et al.*, 2009). Previous work has shown that distributions of species can be accurately predicted using variations in seafloor structure (e.g. Pittman *et al.*, 2007; Dolan *et al.*, 2008; Holmes *et al.*, 2008; Young *et al.*, 2010).

In conjunction with the implementation of the California network of MPAs, ecological surveys using scuba divers were initiated to characterize structure and distribution of populations and biological communities. These surveys were conducted at several MPAs to characterize initial ecological responses to the establishment of MPAs and develop a time series that would facilitate the use of MPAs to inform various marine management issues (e.g. fisheries management, climate change). One limitation of this sampling design, however, is the inability to extrapolate the *in situ* estimates of species densities from samples of limited spatial coverage to estimates of species abundance over broader spatial areas (within and outside MPAs). These extrapolations would allow for predictions of actual abundance and biomass that determine larval production of protected populations as well as distributions of species and assemblages where observations were not made, helping in MPA evaluation.

In this study, we draw from these sources of information and use geographic information systems (GIS), to create SDMs for the purpose of estimating species distribution and abundance at spatial scales relevant to informing adaptive management in the application of a network of MPAs. We use the SDMs to answer the following suite of questions:

1. Do densities of fish species vary predictably with environmental variables including structure of rocky reef, latitude, kelp biomass and exposure to ocean swell (wave orbital velocities)? If so, what is the relative importance of variables

in explaining this variation and how does it vary among species?

2. Do species richness and diversity of kelp forest fish assemblages vary predictably with environmental variables?
3. Can SDMs generate reliable spatial extrapolations of species densities, diversity and richness?
4. Do estimates of fish abundance generated by the SDMs differ substantially (both higher and lower) from those generated by simple uniform extrapolation of density estimates across kelp forest habitat?

Collectively, answers to these questions may demonstrate the utility of SDMs in enhancing our ability to estimate species distributions and abundances and their application for informing spatial and ecosystem-based management approaches.

METHODS

Our general approach to answering these questions was to test for predictive SDMs (generalized additive models) for individual species (Question 1) and the richness and diversity of fish assemblages (Question 2) based on variables extracted from seafloor habitat maps, biogenic structure and oceanographic variables. The SDMs were then used to extrapolate densities, diversity and richness over the study area. We tested the accuracy of these predicted extrapolations using an independent data set (Question 3). We then used the SDMs to estimate species abundances and compared these estimates with other methods for estimating species abundance (Question 4).

Study system

We focus on fish assemblages that inhabit shallow (5–20 m depth) forests of giant kelp, *Macrocystis pyrifera*, along the central coast of California, USA (Fig. 1a), extending from Pigeon Point in the north ($37^{\circ}10'57''$ Lat $122^{\circ}23'38''$ Long) to Point Conception in the south ($34^{\circ}26'55''$ Lat $120^{\circ}28'14''$ Long). This study area corresponds to the Central Coast Study Region of the state Marine Life Protection Act (MLPA) Initiative that created a network of 29 MPAs with differing levels of fishing restrictions (Fig. 1b). The depth range for this study is restricted to 20 m depth due to the limited depth range of the ecological surveys.

We focused analyses on ten species of fishes that were frequently observed in the surveys of the central California kelp forests: tubesnout (*Aulorhynchus flavidus*), black surfperch (*Embiotoca jacksoni*), striped surfperch (*Embiotoca lateralis*), kelp greenling (*Hexagrammos decagrammus*), olive rockfish (*Sebastes serranoides*), kelp rockfish (*Sebastes atrovirens*), gopher rockfish (*Sebastes carnatus*), black rockfish (*Sebastes melanops*), black and yellow rockfish (*Sebastes chrysomelas*) and blue rockfish (*Sebastes mystinus*). With the exception of the tubesnout, all of these species contribute substantially to a culturally and economically important recreational fishery and most are also important constituents of a commercial live-fish fishery (CDFW, 2002a; Starr *et al.*, 2002; Bennett *et al.*, 2004; Stephens *et al.*, 2006a,b; Carr & Reed, 2015). While recreational fisheries have been implicated in declines of nearshore fish stocks (e.g. Love *et al.*, 1998), many of these fisheries are still considered 'data poor' and several lack stock

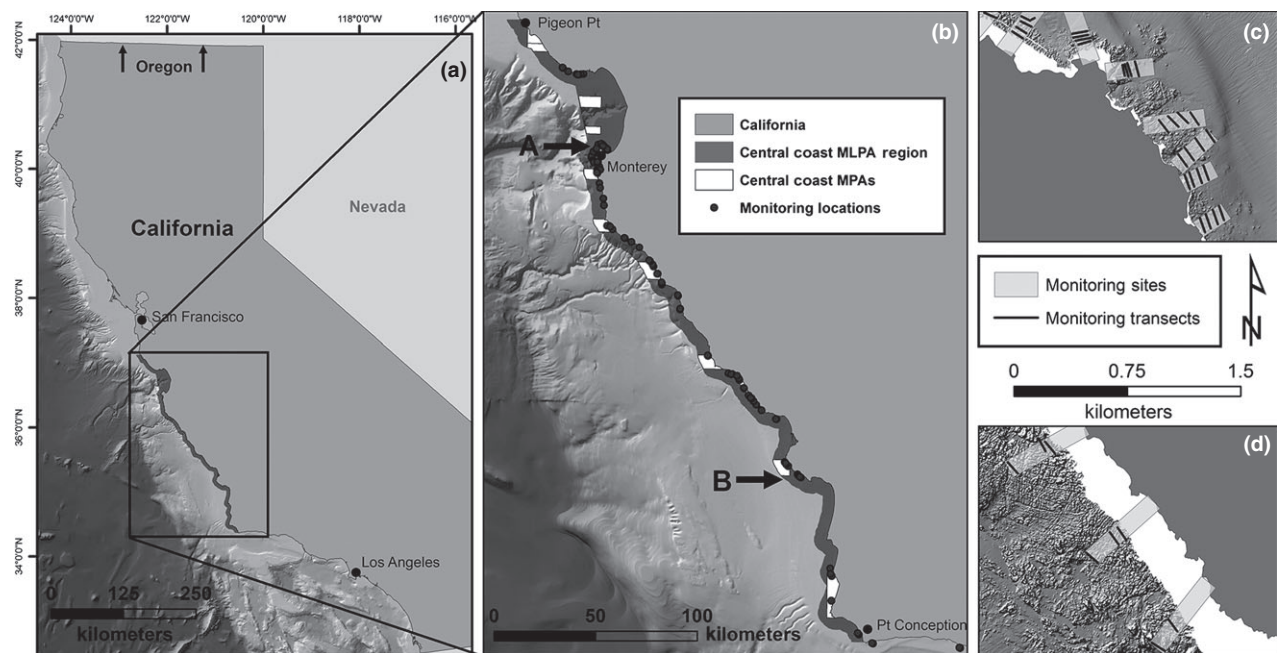


Figure 1 Study location: (a) location of the study area along the central coast of California, USA; (b) central coast Marine Life Protection Act Region indicated in dark grey with marine protected areas across the region shown in white, and points signifying locations of ecological monitoring sites; (c and d) Zoomed in views of the ecological monitoring sites (light grey boxes) with the 100-m survey transects (black lines) overlaid on the 2-m resolution seafloor data.

assessments (CDFW, 2002b). SDMs and predicted density distributions could inform these assessments and spatial management of these fisheries. In addition, species diversity and richness were estimated for the entire assemblage of the 70+ conspicuous species of fishes recorded on ecological surveys.

Estimates of fish densities, diversity and richness

Fish densities were estimated for the ten focal species by visual surveys using scuba, conducted by the Partnership for Interdisciplinary Studies of Coastal Oceans (PISCO: <http://www.piscoweb.org/research/science-by-discipline/eco-system-monitoring/kelp-forest-monitoring/subtidal-sampling-protocol>) at 85 sampling sites distributed across the study area (Fig. 1b). Surveys were conducted from 2007 to 2008 at each site. Three consecutive 30-m-long by 2-m-tall by 2-m-wide (120 m³) transects were sampled in each of four depth zones (5 m, 10 m, 15 m and 20 m) at each sampling site. Transects are stratified from the offshore to onshore margins of the forests both at mid-depth and along the bottom to encompass the 3-dimensional habitat of the kelp forests (Fig. 1c,d). We combined the three transects, including the two portions of the water column, as a single 720-m³ transect as a more conservative estimate of the location of the surveys relative to the seafloor maps (Fig. 1c,d). The locations of these transects were recorded using a GPS unit to mark the location at the beginning of each transect. Only those transects with available position data and overlap with mapped seafloor data were used in this study. From these transects, we derived the mean density of each of the ten species within each depth zone. We also calculated species diversity (H' , Shannon Diversity Index) and richness (number of species) for each transect.

Quantifying habitat characteristics

We used seafloor maps generated by the CSMP to derive a variety of habitat variables that were potentially useful for predicting the density distributions and diversity and richness of the ten fish species. Geomorphic variables from the seafloor included depth, slope, slope of the slope (complexity), topographic position at multiple scales and substrate type (rock versus sediment; Fig. 2). In addition to these variables, we used latitude, wave orbital velocities (a metric of exposure to ocean swell) and kelp biomass estimates (Fig. 2e). The methods used to derive these variables can be found in Appendix S1 in Supporting Information (Geomorphic Variables), Erikson *et al.* (2014; wave orbital velocities) and Cavanaugh *et al.* (2010, 2011; kelp biomass).

We calculated mean and standard deviation of all variables at a 50-m spatial scale except for substrate type using the *Focal Statistics* geoprocessing function in ArcGIS's Spatial Analyst extension. Fifty metres was the finest scale at which predictions could be made with the ecological survey data due to the spatial uncertainty of the sampling locations. For

substrate, we computed the proportion of rocky substrate within each of the buffers.

Do densities of fish species vary predictably with environmental variables?

To test the hypotheses that species densities vary predictably with environmental variables and that the relative importance of these variables differ among species, we used generalized additive models (GAMs). Because response variables in the majority of ecological studies do not have a linear relationship with environmental data, the use of linear regression techniques is often not appropriate (Austin, 1999). Regression techniques including generalized additive models (GAMs), however, allow for nonlinear relationships (Yee & Mitchell, 1991; Austin, 1999). GAMs have become widely used in modelling species distributions (Guisan & Zimmermann, 2000). Therefore, within the 'mgcv' package in R (Wood, 2006; R Development Core Team, 2015), we used GAMs to test for significant predictive relationships between density of each fish species and environmental variables and to identify the relative importance of each environmental variable in contributing to the model. We applied a smoother to those variables in the GAM that did not have a linear relationship with density (Wood, 2006). We combined the total number of transects used in these models across 2007 and 2008 sampling years ($n = 265$), and we developed best-fit models for each species through a process of elimination; variables were removed from the models based on their lack of significance and their collinearity with other variables until the model with the highest deviance explained was produced. We tested for violations of the assumption of independence of observations, including spatial autocorrelation of model residuals (Hurlbert, 1984; Zuur *et al.*, 2009; Zuur *et al.*, 2010).

Do species diversity and richness of the kelp forest fish assemblage vary predictably with environmental variables?

We also used GAMs to test the hypotheses that species richness and diversity vary predictably with environmental variables. We computed species richness (number of species) and the Shannon diversity index (H') for each transect to use as the response variable and used the same environmental variables from the density relationships as independent variables. The total number of transects used in these models was combined across 2007 and 2008 sampling years ($n = 265$). We tested for violations of assumptions of these GAMs as described for the density models.

Can SDMs generate reliable extrapolations of species densities, diversity and richness?

We used the GAM tools within the marine geospatial ecology toolbox (MGET) to create predictive maps extrapolating the density predictions over the study area (Roberts *et al.*, 2010).

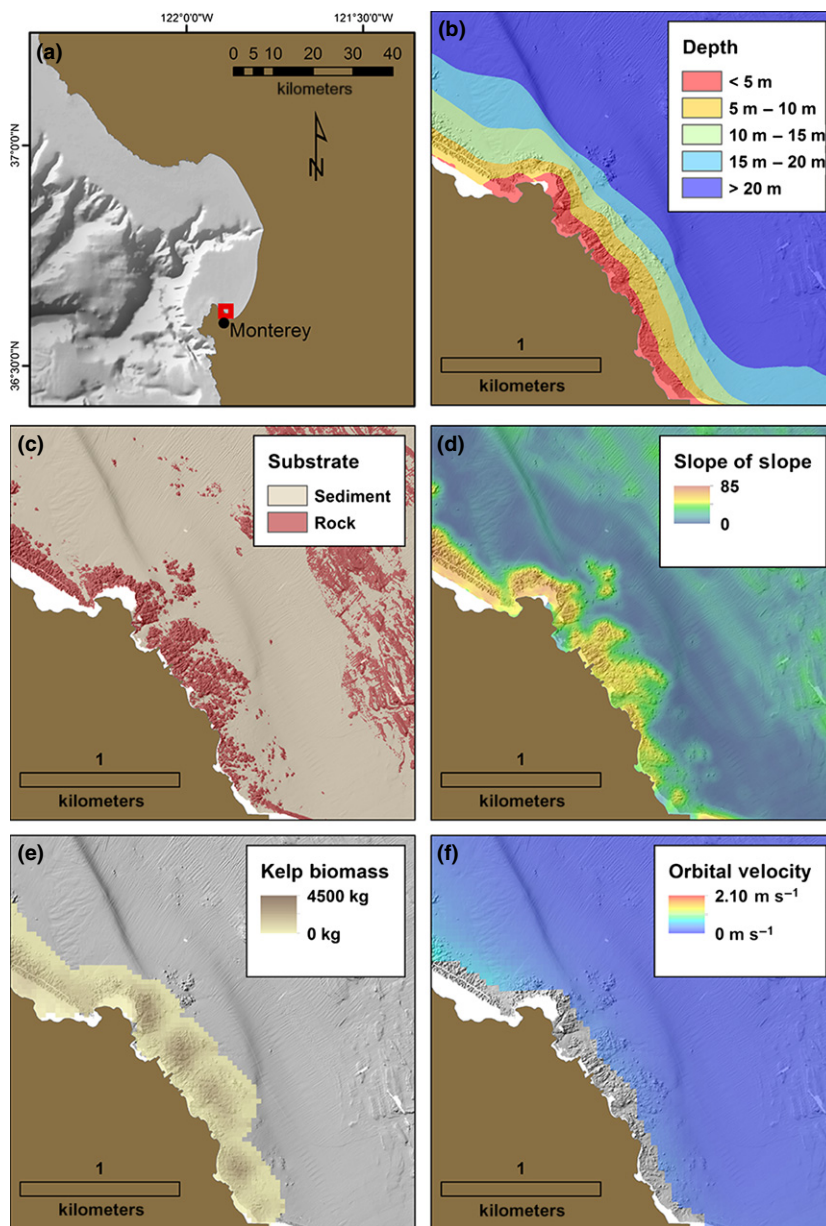


Figure 2 Location of close-up view (red box) of a small portion of the study area on the Monterey peninsula along the central coast of California (a), along with examples of the habitat variables (b–f) used in the species distribution models. Depth strata (b) derived from the 2-m bathymetric DEM with warmer colours representing shallower depths and cooler colours representing deeper depths. Substrate map (c) classified from the VRM analysis with red areas representing ‘rock’ and tan areas representing ‘sediment’. The complexity or maximum rate of maximum slope change between a cell and its eight neighbours (slope of slope, units = degrees of degrees) of the seafloor (d) derived from the 2-m bathymetric DEM with warmer colours representing more complex structure. Distribution of kelp biomass values (e) derived from Landsat data with the darker browns representing greater biomass. Map of wave orbital velocity values (f) with the warmer colours representing higher velocities.

We used 70% ($n = 178$) of the survey transects for these models, reserving the remaining 30% (= 87 transects) to test the accuracy of the density extrapolations. The 87 transects reserved for prediction evaluation were randomly selected and spatially interspersed with transects used to parameterize the SDMs. Because there was no spatial autocorrelation present in the response variables across the sampled sites (see Appendix S2 in Supporting Information), the evaluation transects were independent from transects used to parameterize the SDMs. Random selection of transects allowed us to evaluate the performance of the model representatively across the different spatial regions and habitat types.

We used Pearson’s correlation coefficients to assess the accuracy of density extrapolations of each species by testing for a significant correlation between the density values

predicted by the GAM and the actual density values in the reserved data set. To test whether the Pearson correlation was statistically significant, a Markov chain Monte Carlo (MCMC) simulation was run with 1000 iterations to determine whether the likelihood of these correlations was greater than random chance. We used the same procedure to generate and test the reliability of extrapolated distributions of diversity and richness of the entire sampled fish assemblage.

Comparison of population abundance estimates by different extrapolation methods

For this comparison, we used three methods to extrapolate species densities across the kelp forests in the 0- to 20-m depth zone of one of the MPAs in this region: Point Sur

State Marine Reserve (SMR). The first method, uniform geomorphology, assumed that all rocks are similar. To extrapolate fish density assuming that all rocks are equal, we simply averaged the density of each species across all transects and multiplied that density by the total area of rocky reef identified in the 0- to 20-m depth zone within the MPA. The resulting calculation generates an estimate of the total number (i.e. abundance) of each species in the 0- to 20-m depth zone of the MPA.

The second method used non-spatial geomorphic classifications extracted from the CSMP seafloor maps. The methods for creating these geomorphic classifications can be found in Young & Carr (2015). The six geomorphic classification categories were based on the three-dimensional complexity of the rocky reef: low complexity, low-to-moderate complexity, moderate complexity, moderate-to-high complexity, high complexity and high complexity on sloping terrain. We calculated the product of the density of each species of fish within each geomorphic class and the total area of that geomorphic class within the Point Sur MPA and summed those values across the geomorphic classes.

For the third method, we used the predictive density maps created from the spatially explicit species–habitat associations developed in this study to calculate the total abundance of each species across the region. To do this, we used the density values for each of the 2-m resolution pixels from the

predictive density maps to calculate total abundance within the Point Sur MPA.

RESULTS

Do densities of fish species vary predictably with environmental variables?

The GAMs detected strong correlations between fish density and environmental variables for seven of the ten species analysed. Three species, *A. flavidus*, *H. decagrammus* and *S. mystinus*, did not have strong associations with the habitat variables, and the predictive power of these models was very weak. These three species were not considered in subsequent analyses. Densities of the other seven species showed strong positive or negative correlations with at least three environmental variables (Table 1). Topographic complexity (slope of slope) was the only variable that was consistently significant in all the GAMs. Across all species, the densities had an asymptotic relationship with topographic complexity, and therefore, a smoother was applied to this variable in all models (see Appendix S3 in Supporting Information). Densities of three species, *S. atrovirens*, *S. chrysomelas* and *S. melanops*, were significantly correlated with kelp biomass. *Sebastes atrovirens* and *S. melanops* had a positive, linear relationship with mean kelp biomass. The remaining species, *S. chrysomelas*, had a nonlinear but positive trending relationship with kelp

Table 1 Results of the GAM analyses for the seven kelp forest fishes for which significant density–habitat relationships were detected. Significance of each environmental variable and the deviance explained by the GAM model are presented for each species. Those variables that required a smoothing function are followed by an ‘(s)’ in the environmental variables column

Species	Common name	Environmental variables	Variable significance (P-value)	Deviance explained
<i>Embiotoca jacksoni</i>	Black Perch	Mean Topographic Complexity (s)	<0.001	0.42
		Mean Wave Orbital Velocity (s)	<0.001	
		Mean Depth	0.0229	
<i>Embiotoca lateralis</i>	Striped Perch	Mean Topographic Complexity (s)	<0.001	0.62
		Mean Wave Orbital Velocity	<0.001	
		Latitude	0.038	
<i>Sebastes serranoides</i>	Olive Rockfish	Mean Wave Orbital Velocity (s)	<0.001	0.55
		Latitude (s)	<0.001	
		Mean Topographic Complexity (s)	<0.001	
<i>Sebastes atrovirens</i>	Kelp Rockfish	Mean Wave Orbital Velocity	<0.001	0.51
		Kelp Biomass	<0.001	
		Mean Topographic Complexity (s)	<0.001	
<i>Sebastes carnatus</i>	Gopher Rockfish	Mean Depth (s)	<0.001	0.69
		Latitude	<0.001	
		Mean Topographic Complexity (s)	0.009	
<i>Sebastes chrysomelas</i>	Black and Yellow Rockfish	Mean Depth (s)	<0.001	0.51
		Mean Topographic Complexity (s)	<0.001	
		Kelp Biomass (s)	0.001	
<i>Sebastes melanops</i>	Black Rockfish	Mean Wave Orbital Velocity (s)	0.011	0.48
		Kelp Biomass (s)	<0.001	
		Mean Wave Orbital Velocity (s)	<0.001	
		Mean Topographic Complexity (s)	0.001	

biomass (see Appendix S3). The density of *E. lateralis* and *E. jacksoni* increased with decreasing depth. *Sebastes chrysomelas* and *S. carnatus* had a unimodal relationship with depth; densities tended to be highest at intermediate depths. Densities of three species exhibited negative correlations with exposure to ocean swell (mean wave orbital velocity), and density of only one species, *S. chrysomelas*, was positively correlated with wave orbital velocity. Densities of three species, *E. lateralis*, *S. serranoides* and *S. melanops*, were positively correlated with latitude and only one species, *S. carnatus*, increased in density to the south. The deviance explained (pseudo *R*-squared) by the GAMs generated for these seven species ranged from 0.42 to 0.69 (Table 1).

Do species diversity and richness of the kelp forest fish assemblage vary predictably with environmental variables?

Species diversity (*H'*) and richness were both positively correlated with increases in mean complexity (i.e. slope of slopes; Table 2). Species richness was negatively correlated with mean wave orbital velocity. No other environmental variables contributed significantly to predictive models of species diversity or richness, but the deviance explained for these models was high for both diversity and richness (deviance explained = 0.83).

Can SDMs generate reliable extrapolations of species densities, diversity and richness?

For all but one of the seven species tested, correlation between densities predicted by the GAMs and densities observed in ecological surveys not included in development of the models was strongly significant (Pearson's *r*, $P < 0.001$) (Table 1). We used these relationships to create predictive density distribution maps for each species (e.g. Fig. 3). Predicted densities of *S. serranoides* were not correlated with observed densities (Pearson's $r = 0.26$, $P = 0.35$). Of the significant models, correlation coefficients varied among species from a high of 0.60 ($P < 0.001$) for *S. atrovirens* to a low of 0.40 ($P < 0.001$) for *S. chrysomelas*. Correlations between predicted and observed densities of the two surfperch species, *E. jacksoni* and *E. lateralis*, were similar (Pearson's $r = 0.42$, $P < 0.001$, and $r = 0.46$, $P < 0.001$, respectively). GAMs for the remaining two species, *S. carnatus* and *S. melanops*, were 0.55 ($P < 0.001$) and 0.44 ($P < 0.001$),

respectively. Thus, the average accuracy of the models based on the Pearson's correlation coefficient across all the models was 0.45. Using the same validation approach, the GAMs were even better at predicting spatial patterns of diversity ($r = 0.61$; $P < 0.001$) and richness ($r = 0.71$; $P < 0.001$) of the entire fish assemblage sampled across the study area.

Effect of different methods for estimating population abundances within the central coast MPA network

The comparison between the three methods for calculating species abundances (uniform geomorphology, non-spatial geomorphic and SDM) indicates that density extrapolations assuming uniform geomorphology produce much greater estimates of species abundance than the non-spatial variable geomorphology and spatially explicit environment-based estimates (Table 3). The magnitude of the differences in estimated abundance varied markedly among the seven species, ranging from two to 17 times greater abundance estimates. With the exception of *E. jacksoni*, the abundance estimates from the SDM calculations produced the lowest predicted abundance of the three methods. Overall, the non-spatial geomorphic and spatially explicit SDM-based extrapolations were more similar to each other than either predictions were to uniform geomorphic extrapolation (Table 3).

DISCUSSION

Most of the temperate, nearshore fishes examined in this study exhibited strong associations with environmental variables that constitute their habitat, which allowed for accurate extrapolation of their densities within kelp forests across the study region. Quantitative relationships between fish density and environmental variables also made it possible to better understand potential sources of variation in the distribution of seven of the ten species originally targeted in the study. Across all seven species, species density was correlated with three to four variables, indicating that multiple aspects of the environment must be considered when explaining spatial variation and mapping species distributions.

The three-dimensional complexity (topographic complexity) of the rocky habitat (slope of slope) was the only variable that was significant across all seven species modelled, species richness and diversity. The relationship between complexity and all the response variables was asymptotic. Species

Table 2 Results of the GAM analyses for the fish assemblage attributes. Significance of each environmental variable and the deviance explained by the GAM model are presented for each assemblage attribute. Those variables that required a smoothing function are followed by an '(s)' in the environmental variables column

Assemblage attribute	Environmental variables	Variable significance (<i>P</i> -value)	Deviance explained
Species Diversity (Shannon–Weiner)	Mean Topographic Complexity (s)	<0.001	0.83
Species Richness (no. of species)	Mean Topographic Complexity (s)	<0.001	0.83
	Mean Orbital Velocity (s)	<0.001	

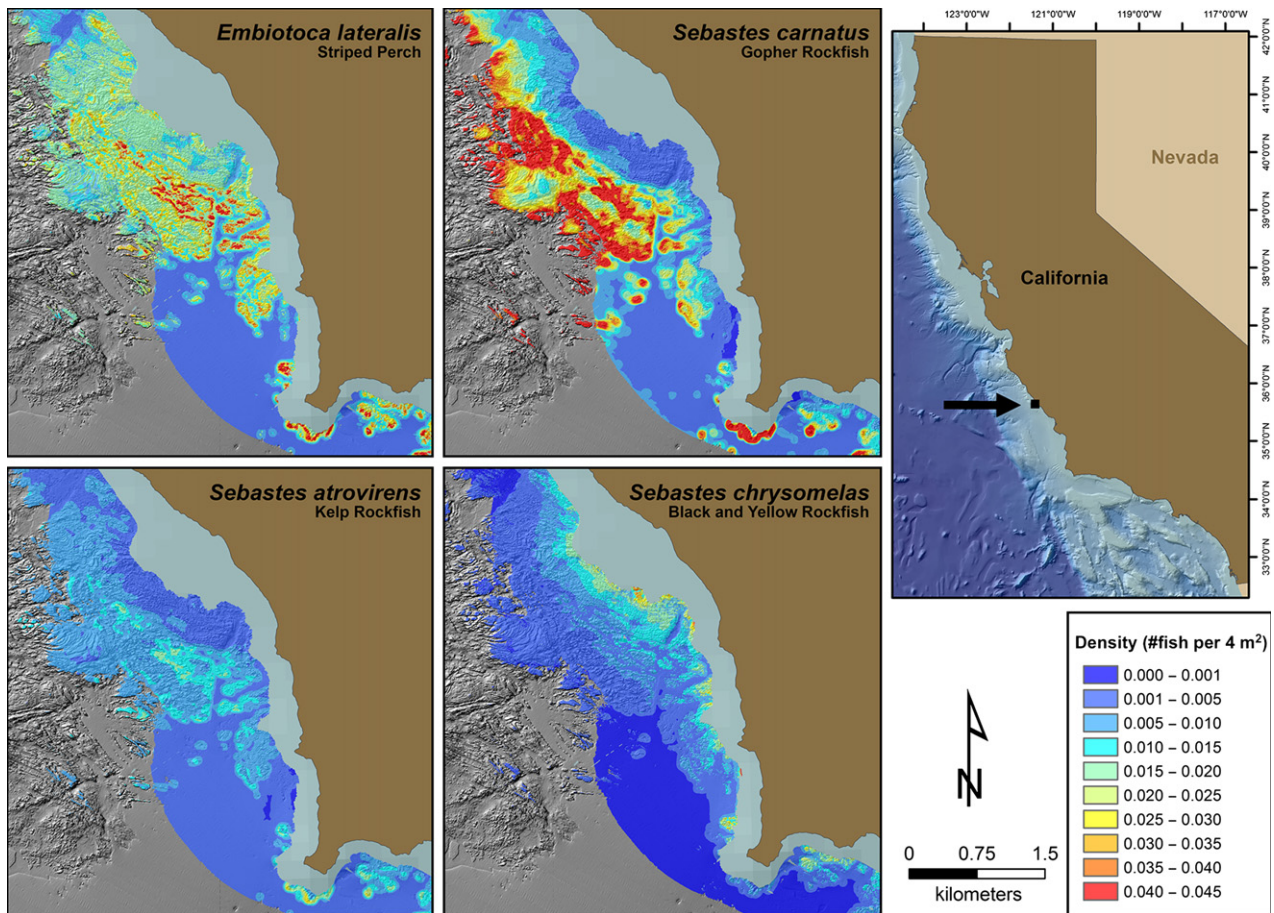


Figure 3 Examples of the species density distribution maps created for four of the seven species successfully modelled in this study. The warmer colours represent higher densities. The variation in density distribution for these four species can clearly be seen in this one small area within the Central Coast Marine Life Protection Act region. Brown indicates land mass and the adjacent light grey area is the shallow subtidal zone in which the seafloor could not be mapped.

densities, diversity and richness all increased with increasing complexity up to slope of slope values between 10 and 15. None of these variables increased with further increases in complexity. This result implies that complexity values associated with suitable habitat begin between slope of slope values of 10 to 15 and any greater value indicates suitable habitat. The large and consistent contribution of structural complexity of the reef in explaining variation in fish densities is consistent with other species distribution models generated for coral reef fishes (Beger *et al.*, 2003; Pittman *et al.*, 2009; Kendall & Miller, 2010; Pittman & Brown, 2011). Thus, heterogeneity of the rocky substratum is a key correlate with the distribution of many demersal species and a likely critical component of any predictive species distribution model.

The proportion of rock was not a significant variable in any of the GAMs for the species successfully modelled. During the design of the network of MPAs along the central coast of California, 'rock' and 'sediment' were the only substrate variables considered (Gleason *et al.*, 2013; Saarman *et al.*, 2013). The large contribution of topographic complexity in explaining variation in fish density of the species

examined here indicates that consideration of rock substratum alone without recognition of topographic complexity can fail to identify important patterns of species abundance, indicating that spatial and ecosystem-based management should strive to identify and consider patterns of topographic complexity (Moore *et al.*, 2011; Rees *et al.*, 2014).

Correlations between species density and depth reflected the relative spatial scale of the ecological surveys and the depth ranges of the species. First, depth contributed to models only for those species whose depth range did not span the entire depth range of the surveys. The distribution of two species pairs, *E. lateralis* and *E. jacksoni*, and *S. chrysomelas* and *S. carnatus*, is depth-stratified, reflecting interspecific competition within these pairs. *E. lateralis* and *S. chrysomelas* inhabit shallower depths relative to their respective congeners within the depth range sampled (Hixon, 1980; Larson, 1980a,b; Schmitt & Holbrook, 1990). Nonetheless, for both species of surf perches (genus *Embiotoca*) and *S. chrysomelas*, density was more strongly correlated with variables other than depth. Only for *S. carnatus* was depth the strongest correlate with density. These results suggest that

Table 3 Comparison of species abundance estimates generated by three methods for extrapolating species density within the Point Sur MPA: uniform extrapolation treating all rocks as equal, non-spatial habitat-based extrapolation and the abundances predicted from the spatially explicit species distribution models

Species	Common name	Uniform extrapolated abundance	Geomorphic-based extrapolated abundance	SDM-based extrapolated abundance
<i>Embiotoca jacksoni</i>	Black Perch	9149	2897	4890
<i>Embiotoca lateralis</i>	Striped Perch	59,065	23,014	22,655
<i>Sebastes serranoides</i>	Olive Rockfish	157,071	46,466	19,895
<i>Sebastes atrovirens</i>	Kelp Rockfish	38,133	13,313	9198
<i>Sebastes carnatus</i>	Gopher Rockfish	69,650	19,072	14,621
<i>Sebastes chrysomelas</i>	Black & Yellow Rockfish	20,977	11,315	10,817
<i>Sebastes melanops</i>	Black Rockfish	161,165	12,844	8666

within the depth range of kelp forests (5–20 m depth), other variables are important correlates with density. Depth ranges of all of these species span and extend beyond the depth range of the surveys. To better understand the relative contribution of depth to species–habitat relationships for these species, observations across their entire depth range would need to be incorporated in the models.

Latitude was an important variable in helping to explain the distribution of *E. lateralis*, *S. carnatus*, *S. melanops* and *S. serranoides*, and the distribution and abundance of rocky reef fishes is known to vary with latitude along the coast of California (McClatchie *et al.*, 1997; Stephens *et al.*, 2006a,b; Carr & Reed, 2015). Of these four species, *S. carnatus* was the only one whose density decreased to the north along the central coast of California. These correlations with latitude likely reflect relationships with other environmental variables that covary with latitude along the coast including temperature, upwelling, exposure to ocean swell and the persistence of kelp forests (Reed *et al.*, 2011; Carr & Reed, 2015).

Biomass of *Macrocystis pyrifera* proved to be an important variable, positively correlated with the density of three species, and was the most important variable in explaining the distribution of *S. chrysomelas*. These results add to a number of other studies that have identified correlations between fish density and the presence or density of *M. pyrifera* (reviewed by Stephens *et al.*, 2006a,b; Carr & Reed, 2015).

Wave orbital velocities had a negative relationship with the density of *E. jacksoni*, *S. serranoides* and *S. atrovirens* and a positive relationship with *S. chrysomelas*. These limited results show that features of the water column are important correlates with the distribution and abundance of shallow dwelling fish species. Incorporation of features of the water column in species distribution models for shallow benthic species will likely increase their explanatory power and capacity to predict species distributions across nearshore marine landscapes (Hinchey *et al.*, 2008).

Although the majority of the species showed strong habitat affinities, some of the species were not successfully modelled in this study. *Aulorhynchus flavidus*, *H. decagrammus* and *S. mystinus* all failed to show any strong relationship

with the habitat variables used. The reason for the lack of significance of variables could be due to the number of observations of each of these species. *Hexagrammos decagrammus* and *A. flavidus* both had fewer observations than the other species examined in this study possibly resulting in failure of the model to find any significant relationship with habitat variables. On the other hand, *S. mystinus* had a large number of observations but were found over a wide variety of habitats and appeared to be more of a habitat generalist. Additionally, *S. mystinus* and *A. flavidus* aggregate in the water column well above the reef surface. This behaviour may have contributed to the lack of any significant relationship with finer scale habitat variables associated with the reef.

Models that predict spatial variation in attributes of the fish assemblage (species diversity and richness) are much simpler than those for the individual species. Both richness and diversity are positively and strongly associated with the structural complexity of the rocky reef. Fish species richness is also negatively related to wave orbital velocity. These assemblage-level models are most likely simpler than the species-specific models because of the variety of fish species examined, which collectively exhibited more general habitat associations than the individual species.

It is important to recognize the limitations of these models to extrapolate densities and estimate species abundances. Such estimates are only appropriate over areas that have been subsampled to parameterize the model. This underscores the importance of identifying the area of interest, spatially distributing ecological surveys accordingly, testing the accuracy of predictions within that area and restricting density and abundance estimates to the study area.

Methods for evaluating performance of a predictive model are widely debated throughout the literature (Araújo & Guisan, 2006). Although there are proponents for evaluating the predictive performance of models using completely independent data sets, the use of pre-existing data sets, which were not collected for the purpose of creating SDMs, often precludes that possibility. Therefore, using a random subset of data set aside for model evaluation becomes a necessary option. Because our observations were not spatially

autocorrelated, the reserved data provided us with observations independent of the training data, which is a requirement of evaluation data (Araújo *et al.*, 2005). The random subsampling of transects also allowed us to have observations spatially interspersed with the training data, so we could evaluate the model performance over a variety of habitats and regions.

The comparison between the different methods for calculating abundance for the six species successfully modelled with the SDMs showed that incorporating habitat heterogeneity produces substantially lower estimates of abundance. In addition, abundance estimates generated by the non-spatial and SDM derived estimates were closer to one another than those derived from the uniform extrapolations. These results warn that, by not accounting for habitat variation, calculations of abundance could greatly overestimate population sizes.

Taken together, our results demonstrate the utility of SDMs for identifying environmental variables that likely influence spatial patterns of species densities and assemblage attributes (diversity, richness). These relationships can inform the design of spatially explicit conservation approaches such as protected areas by ensuring critical habitat is included within proposed protected areas and estimate population size of species under protection. Incorporating size structure can also be used to estimate larval production and the contribution of protected populations to the replenishment of populations within and outside of protected areas. Thus, spatially explicit patterns of abundance and biomass underpin regional population models that can be used to evaluate the design of MPA networks, like that along the coast of California, and inform adaptive management of networks to protect and enhance regional populations. Similarly, spatial patterns of species richness or diversity of assemblages can inform spatial approaches to conservation designed to protect biodiversity.

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SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article:

Appendix S1 Seafloor geomorphology variables.

Appendix S2 Spatial autocorrelation analyses.

Appendix S3 Relationships between response and explanatory variables.

BIOSKETCHES

Mary Young is a postdoctoral researcher at Deakin University in Warrnambool, Australia, who recently finished her Ph.D. in the Ecology and Evolutionary Biology Department at University of California at Santa Cruz. In her research, she applies methods from landscape ecology to understand ecological patterns in the marine environment including species distributions, population connectivity and population persistence. Mary also uses these landscape ecology techniques for management applications such as the evaluation of marine protected area placement.

Mark H. Carr is a professor in the Department of Ecology and Evolutionary Biology at the University of California at Santa Cruz (<http://research.pbsci.ucsc.edu/eeb/rclab/>). He and his graduate students study the population and community ecology of coastal marine and anadromous fishes, and coastal marine ecosystems. Dr Carr's ecological research informs management and conservation topics including ecosystem-based fisheries management, design and evaluation of marine protected areas, and large-scale, long-term monitoring studies.

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